Soldering Iron Safety Policy

1. Purpose

This policy defines the mandatory safety rules for using soldering irons, soldering stations, hot tweezers, desoldering tools, and related bench soldering equipment in the lab. It is intended to prevent burns, fire, electrical shock, fume exposure, chemical exposure, eye injury, cuts, and property damage.

Every employee, contractor, intern, visitor, or other person who solders, assists with soldering, cleans soldering equipment, or works near active soldering work must follow this policy and sign the acknowledgement before participating in soldering work.

2. Scope

This policy applies to:

- Hand soldering, rework, desoldering, wire tinning, connector soldering, PCB assembly, and PCB repair.
- Use of soldering irons, soldering stations, hot tweezers, desoldering irons, solder suckers, solder wick, flux, tip cleaners, fume extraction, and related hand tools.
- Handling of hot components, molten solder, flux, cleaning chemicals, cut leads, wires, PCBs, and solder waste.
- Anyone working near the soldering area while tools are powered, heating, in use, or cooling.

This policy does not cover reflow ovens, hot plates, hot air rework stations, laser work, chemical etching, battery welding, or other higher-energy thermal processes, which require separate procedures.

3. Conservative Safety Assumption

The person handling soldering equipment shall treat every soldering iron, tip, stand, hot tweezer, desoldering nozzle, recently soldered component, and recently soldered PCB as hot unless they have personally verified that it is cool.

The operator shall treat soldering work as a combined heat, electrical, chemical, fume, sharp-object, and fire hazard. Familiarity with soldering does not remove the need for ventilation, PPE, housekeeping, and active attention.

If any tool is damaged, unstable, producing smoke from the handle or cord, sparking, overheating abnormally, failing to regulate temperature, or missing its stand, it must be removed from service until inspected and cleared.

4. Main Ways a Person Could Be Hurt

Employees must understand that injury can occur during routine soldering, setup, cleanup, rework, and tool handling. The principal hazards are:

1. **Burns from hot tools.** Soldering iron tips, hot tweezers, desoldering nozzles, stands, recently soldered joints, components, wires, and PCBs can burn skin.
2. **Burns from molten solder.** Molten solder can splash or drip onto skin, eyes, clothing, the bench, or nearby equipment.
3. **Eye injury.** Solder splatter, clipped component leads, snapped wire, flux spatter, solder wick fragments, and tool slips can injure eyes.
4. **Fume exposure.** Flux fumes, solder smoke, heated residues, adhesives, conformal coating, wire insulation, and contaminated boards can irritate the eyes, nose, throat, and lungs.
5. **Lead or metal exposure.** Leaded solder, solder dust, dross, contaminated wipes, and dirty hands can expose users to lead or other metals if ingested or transferred to food, drinks, phones, or personal items.
6. **Chemical exposure.** Flux, isopropyl alcohol, cleaning solvents, tip tinner, adhesives, conformal coating removers, and residues can irritate skin, eyes, and lungs.
7. **Fire.** Hot tools can ignite paper, cardboard, wipes, packaging, dust, solvents, aerosols, foam, plastic bags, clothing, or other combustible materials.
8. **Electrical shock.** Damaged cords, wet work surfaces, improper grounding, exposed wiring, defective power supplies, or energized circuits can create shock hazards.
9. **Cuts and punctures.** Component leads, wire ends, PCB edges, tweezers, cutters, blades, and broken parts can cut or puncture skin.
10. **Unexpected energized circuits.** Soldering on powered equipment, charged capacitors, batteries, or unknown circuits can cause shock, sparks, fire, component rupture, or tool damage.
11. **Ergonomic strain.** Poor posture, repetitive fine work, awkward wrist position, excessive force, and long soldering sessions can cause strain or fatigue.
12. **Complacency risk.** Injuries are more likely when an operator leaves an iron loose on the bench, skips fume extraction, eats or drinks near soldering work, rushes, or assumes a tool has cooled.

5. Operator Responsibilities

5.1 Person Handling Soldering Equipment

The person handling soldering equipment shall:

- Complete required training before use.
- Follow this policy, the manufacturer's instructions, and the lab operating procedure.
- Confirm the work area, ventilation, soldering station, stand, power cord, tip, solder, flux, and board are safe before use.
- Wear required PPE.
- Use fume extraction or local ventilation during soldering.
- Keep the hot tool in its stand whenever it is not actively in hand.
- Keep combustible materials and liquids away from hot tools.
- Stop work immediately if unsafe conditions exist.
- Report injuries, near misses, fire, smoke events, electrical faults, damaged cords, missing stands, unstable stations, or ventilation failures.
- Remove damaged or unsafe equipment from service.

5.2 Observers and Visitors

Observers and visitors may approach the soldering area only with permission from the person handling the equipment. They must avoid hot tools, fumes, molten solder, sharp leads, and active work areas. Visitors may not solder unless trained and authorized.

6. Authorization and Training

No person may solder independently until all of the following are complete:

- Review of this policy and signature of the acknowledgement.
- Review of the current lab soldering procedure.
- Hands-on training by a person already competent with the soldering tools and process.
- Demonstrated ability to set up the station safely, use fume extraction, select suitable solder and flux, solder without creating unsafe conditions, place the iron in the stand, clean the tip, handle hot parts, clean the bench, and respond to burns, smoke, or fire.
- Demonstrated understanding of hot surfaces, fume hazards, lead hygiene, chemical handling, fire prevention, electrical hazards, required PPE, and incident reporting.

Refresher review is required after any incident, near miss, equipment change, process change, or extended period without soldering.

7. Approved Materials and Ventilation

Only approved solder, flux, wire, components, PCBs, cleaning agents, and rework materials may be used.

Fume extraction or local ventilation must be used whenever soldering, desoldering, heating flux, burning off residue, or working on contaminated boards. The extractor inlet must be positioned close enough to capture fumes before they enter the breathing zone.

The following are prohibited unless specifically approved by the person responsible for the lab or equipment:

- Soldering unknown, contaminated, wet, oily, chemically treated, or visibly degraded boards.
- Heating unknown coatings, plastics, adhesives, conformal coatings, potting compounds, or insulation.
- Soldering materials that produce strong fumes, smoke, charring, or unknown decomposition products.
- Using unapproved solvents, acids, corrosive fluxes, or aggressive cleaners.
- Using leaded solder without approved lead-handling controls.
- Soldering on powered circuits, batteries, or charged capacitors unless covered by a specific safe procedure.

If a material produces strong odor, visible smoke beyond normal flux fumes, charring, bubbling, or unexpected reaction, stop work and report the issue.

8. Required Engineering Controls

The following controls must be in place before soldering:

- A stable soldering station with temperature control where required.
- A proper soldering iron stand that securely holds the hot tool.
- A nonflammable or heat-resistant work surface.
- Fume extraction or local ventilation.
- Adequate lighting and a clear bench.
- Safe cable routing so cords cannot pull the iron off the bench.
- A suitable fire extinguisher accessible in the lab.
- Approved containers for solder waste, clipped leads, used solder wick, contaminated wipes, and chemical waste.
- Labeled chemical containers with lids closed when not in active use.

If any required control is missing or defective, soldering must not begin.

9. Required PPE

Minimum PPE for normal soldering:

- Safety glasses.
- Heat-resistant finger protection or gloves when handling hot parts, large thermal-mass components, or recently soldered assemblies.

- Nitrile gloves when handling flux, solvents, contaminated boards, leaded solder residue, or cleaning chemicals.
- Cut-resistant gloves when handling sharp sheet metal, sharp PCBs, or sharp assemblies if they do not create an entanglement or dexterity hazard.

Operators must secure long hair, loose clothing, jewelry, lanyards, hoodie strings, and dangling items before soldering.

PPE does not make it safe to touch a hot soldering tip, molten solder, energized circuit, or unventilated fumes.

10. Pre-Use Checklist

Before soldering, the operator shall:

1. Confirm they are trained and not impaired, distracted, or rushed.
2. Confirm the bench is clean, stable, dry, and free of unnecessary combustible material.
3. Confirm the soldering station, stand, tip, cord, plug, power supply, and temperature control are undamaged.
4. Confirm fume extraction or local ventilation is operating and positioned correctly.
5. Confirm the workpiece is unpowered, disconnected, and safe to solder.
6. Discharge capacitors where applicable using an approved method.
7. Confirm batteries are removed or protected under a specific safe procedure.
8. Confirm solder, flux, cleaners, and materials are approved for the task.
9. Confirm safety glasses and other required PPE are worn.
10. Confirm the fire extinguisher is accessible.

If anything is uncertain, stop and do not solder until the safe setup and procedure are clear.

11. Operating Rules

Operators shall follow these rules during every soldering task:

- Keep the soldering iron in its stand whenever it is not actively in hand.
- Never place a hot iron directly on the bench.
- Never point or pass a hot iron toward another person.
- Keep hands, wires, sleeves, cords, and tools clear of the hot tip.
- Use fume extraction continuously while soldering.
- Do not inhale fumes intentionally or lean directly over the solder joint.
- Do not flick solder from the tip.
- Do not eat, drink, vape, apply cosmetics, or handle contact lenses in the soldering area.
- Wash hands after soldering, especially before eating, drinking, or touching personal items.
- Do not use compressed air to blow solder dust, clipped leads, or residues into the room.

- Do not solder on powered circuits unless a specific safe procedure allows it.
- Do not leave a powered soldering iron unattended.
- Do not use damaged cords, cracked handles, unstable stands, loose tips, or malfunctioning stations.
- Turn down, sleep, or power off the station during long pauses according to the lab procedure.

Stop immediately if the tool behaves unexpectedly, fumes are not being captured, smoke appears from the handle or cord, the board chars, or the workpiece becomes unstable.

12. Lead and Hygiene Controls

If leaded solder is used or may be present:

- Treat solder, dross, used solder wick, dirty tips, contaminated wipes, and bench residue as lead-contaminated.
- Keep lead-containing materials labeled and separated where required.
- Wash hands after soldering and before eating, drinking, or leaving the lab.
- Keep food, drinks, gum, cosmetics, and personal items away from the soldering area.
- Do not sand, grind, or abrade solder joints unless a specific dust-control procedure is used.
- Dispose of lead-contaminated waste according to lab waste procedures.

Lead-free solder still requires fume extraction, hand washing, and good housekeeping.

13. Shutdown and Post-Use Procedure

After soldering, the operator shall:

1. Place the iron securely in its stand.
2. Clean and tin the tip according to the lab procedure.
3. Power off the soldering station unless another trained operator is immediately taking over.
4. Allow the tip, stand, parts, and PCB to cool before handling or storage.
5. Return solder, flux, tools, and chemicals to their proper storage locations.
6. Dispose of solder waste, clipped leads, used solder wick, contaminated wipes, and chemical waste properly.
7. Clean the bench without spreading dust or residue into the room.
8. Wash hands after cleanup.

Do not store a hot iron, hot PCB, or hot component in a drawer, bin, bag, or near combustible material.

14. Maintenance and Troubleshooting

Routine maintenance may be performed only by trained users. Non-routine maintenance, electrical repair, station disassembly, calibration, or cord replacement must be performed only by a person competent for that task under a specific safe procedure.

Before maintenance, the operator shall:

- Power off and unplug the station when required.
- Allow hot components to cool.
- Use proper tools and PPE.
- Keep liquids away from electrical parts.

During troubleshooting:

- Never touch exposed energized parts.
- Never use a station with a damaged cord, cracked handle, loose plug, unstable stand, or suspected electrical fault.
- Never bypass grounding, fuses, temperature control, or safety features.
- Tag unsafe equipment out of service and report it.

15. Housekeeping and Waste

The soldering area must be kept clean and orderly.

Operators shall:

- Keep the bench free of loose paper, cardboard, packaging, wipes, foam, plastic bags, and unnecessary combustibles.
- Keep sharp leads and wire clippings contained.
- Keep solder, flux, cleaners, and solvents capped and labeled.
- Keep wet items and liquids away from powered equipment.
- Dispose of solder waste, lead-containing waste, contaminated wipes, used solder wick, and chemical waste according to lab procedures.
- Clean residues using approved methods.

16. Emergency Procedures

16.1 Burns

- Move away from the hot source.
- Cool the burn with clean cool water according to first-aid procedure.
- Do not pull hardened solder or plastic from skin if it is stuck.
- Report the injury immediately.
- Seek medical attention for serious burns, burns involving molten solder, electrical burns, or burns larger than a minor superficial injury.

16.2 Eye Injury

- Stop work immediately.
- Do not rub the eye.
- Use the eyewash station if chemical exposure or particles are suspected and it is appropriate to do so.
- Seek medical attention for solder splatter, clipped lead impact, chemical splash, embedded particles, or persistent irritation.
- Report the incident immediately.

16.3 Cuts or Punctures

- Stop work.
- Use first aid as appropriate.
- Seek medical attention for deep cuts, punctures from contaminated leads, embedded debris, or uncontrolled bleeding.
- Report the injury immediately.

16.4 Fume Exposure or Ventilation Failure

- Stop soldering immediately.
- Move away from the fumes.
- Turn off the soldering station if safe to do so.
- Notify the person responsible for the lab or equipment.
- Do not resume until ventilation is restored and the cause is understood.
- Seek medical attention for breathing difficulty, chest tightness, severe irritation, dizziness, or persistent symptoms.

16.5 Fire or Smoke

- Stop work immediately.
- Power off or unplug the soldering station only if safe to do so.
- Alert nearby personnel.
- Use a fire extinguisher only if trained, the fire is small, and there is a clear exit path.
- Evacuate and call emergency services if the fire grows, produces heavy smoke, involves electrical equipment, or cannot be controlled immediately.
- Do not resume operation until the equipment and work area have been inspected and the cause has been corrected.

16.6 Electrical Fault

- Stop work.
- Do not touch exposed wiring, damaged cords, wet equipment, or energized parts.

- Disconnect power only if safe to do so.
- Tag the equipment out of service.
- Notify the person responsible for the lab or equipment.

17. Stop-Work Authority

Every employee has authority to stop soldering work if they believe conditions are unsafe. Work may resume only when the unsafe condition is corrected and the person handling the soldering equipment has confirmed the setup is safe.

18. Employee Acknowledgement

By signing below, I acknowledge that:

- I have read and understood this Soldering Iron Safety Policy.
- I understand the ways soldering can injure me or others, including burns, molten solder splatter, eye injury, fumes, lead or metal exposure, chemical exposure, fire, electrical shock, cuts, punctures, and ergonomic strain.
- I agree to solder only when trained, unimpaired, and following the safe procedure.
- I agree not to leave hot tools unattended, bypass safety controls, solder on unsafe circuits, use damaged equipment, skip fume extraction, or allow untrained operation.
- I agree to report incidents, near misses, equipment faults, unsafe conditions, smoke, fire, abnormal odors, and injuries immediately.

Employee name: ______________________________

Employee signature: ___________________________

Date: _______________________________________

19. References for Safety Basis

- Manufacturer manuals, labels, specifications, and maintenance instructions for the soldering irons, soldering stations, fume extractors, and related tools used in the lab.
- Safety data sheets and supplier information for solder, flux, cleaners, solvents, adhesives, conformal coatings, and other soldering materials.
- OSHA general industry safety principles for personal protective equipment, hazard communication, electrical safety, fire prevention, and safe work practices.
- NIOSH and other occupational health guidance regarding soldering fumes, flux fumes, lead exposure, and local exhaust ventilation.

Signature

Printed Name	Signature
Date	Supervisor / Reviewer